

# EXP34 Morphological operations based on OpenCV using Top hat technique.

AIM:- Morphological operations using Top hat technique

## PROGRAM:

```
import cv2

filterSize = (3, 3)

kernel = cv2.getStructuringElement(cv2.MORPH_RECT, filterSize)

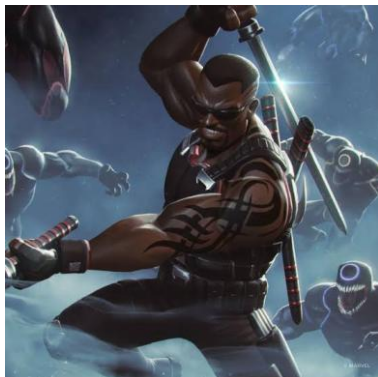
input_image = cv2.imread("C:\\Users\\pavan\\OneDrive\\Opencv\\blade.jpg")
input_image = cv2.cvtColor(input_image, cv2.COLOR_BGR2GRAY)
tophat_img = cv2.morphologyEx(input_image, cv2.MORPH_TOPHAT, kernel)
cv2.imwrite("tophat.jpg", tophat_img)

cv2.imshow("Original Image", input_image)
cv2.imshow("Top Hat Image", tophat_img)

cv2.waitKey(0)

cv2.destroyAllWindows()
```

## INPUT:



## OUTPUT:



**RESULT:-** Thus the program is executed successfully.